Ain Shams University

Faculty of Engineering

CHEP – Communications Systems Engineering Program

Spring 2026 | ECE359: Digital Image Processing & Multimedia

Sheet 2

Pixel Relationships, Image Operations & Spatial Transformations

Part A Textbook Problems

1. [1] Consider the two image subsets S_1 and S_2 shown in the figure below.

0	0	1	0	0
0	1	1	0	0
0	0	1	0	0
0	0	1	1	0

where S_1 contains the upper cluster and S_2 the lower.

For $V = \{1\}$, determine whether these two subsets are:

(a) 4-adjacent,
(b) 8-adjacent,
(c) m-adjacent.

Justify each answer by identifying the specific pixel pair(s) that establish or deny the adjacency, and state clearly which pixel coordinates are involved.

2. [2] Consider the following 4×4 image segment, where pixel p is located at the **bottom-left** and pixel q at the **top-right**:

3	1	2	$1_{(q)}$
2	2	0	2
1	2	1	1
$1_{(p)}$	0	1	2

(a) Let $V = \{0, 1\}$. Compute the lengths of the shortest 4-, 8-, and m-path between p and q. If a particular path does not exist, explain why.
(b) Repeat part (a) for $V = \{1, 2\}$.

[1] R.C. Gonzalez & R.E. Woods, *Digital Image Processing*, 3rd ed. (Prentice Hall, 2008), Problem 2.11.
[2] Gonzalez & Woods, 3rd ed., Problem 2.15.

3. [3] Show that **image averaging can be performed recursively**. Specifically, let $\bar{g}(k)$ denote the running average of k images. Show that the average of $k+1$ images, $\bar{g}(k+1)$, can be obtained directly from $\bar{g}(k)$ and the newly acquired image f_{k+1}, without re-summing all previous images. Derive the recursive update formula and state any assumptions made about the noise model.

4. [4] With reference to the standard affine transformation matrices in Table 2.3, derive **single, composite transformation matrices** for each of the following compound operations. Express each result as a single 3×3 homogeneous matrix.

(a) Scaling followed by translation.
(b) Scaling, followed by translation, followed by rotation.
(c) Vertical shear, followed by scaling, followed by translation, followed by rotation.
(d) Does the *order* in which the individual matrices are multiplied affect the final composite transformation? Justify your answer with a concrete numerical example involving a scaling/translation pair.

5. [5] The general affine transformation is written as:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

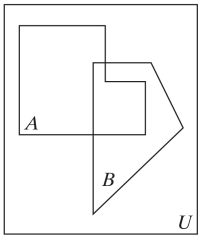
Derive the **inverse transformation** $\mathbf{A}^{-1}$ (i.e. the mapping from output coordinates back to input coordinates) for each of the following:

(a) Scaling.
(b) Translation.
(c) Vertical and horizontal shearing.
(d) Rotation.
(e) A composite translation followed by rotation (derive a single inverse matrix).

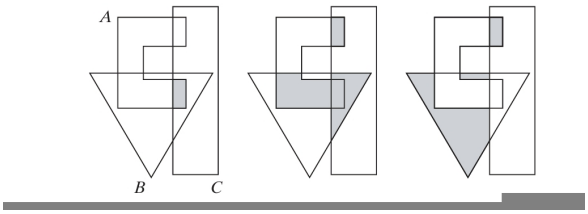
6. [6] With reference to the standard set-theoretic operations and Venn diagram notation:

(a) Sketch the set $(A \cap B) \cup (A \cup B)^c$.
(b) For each of the three shaded Venn diagrams provided in Fig. 2.31 of the textbook, write a single set expression in terms of sets A, B, and C that exactly describes the shaded region. Each shaded area constitutes one set; give one expression per figure.

[3] Gonzalez & Woods, 4th ed., Problem 2.25.
[4] Gonzalez & Woods, 4th ed., Example/Exercise 2.36. Refer to Table 2.3 (Affine Transformation Matrices) in the textbook.
[5] Gonzalez & Woods, 4th ed., Example/Exercise 2.37.
[6] Gonzalez & Woods, 3rd ed., Problem 2.23.



Question 6 (a)



Question 6 (b)

Part B Section Problems

7. (m-Adjacency Analysis)
Consider the following 2×2 pixel neighbourhood, where the intensity values are labelled by their pixel identifiers. Let $V = \{1\}$.

$1_{(p)}$	$1_{(A)}$
$0_{(B)}$	$1_{(q)}$

(a) Determine whether pixels p and q are **8-connected**. Identify the type of neighbour relationship between them.
(b) Determine whether pixels p and q are **m-connected**. Apply both rules of m-adjacency formally:
- Rule 1 (4-neighbour condition), and
- Rule 2 (diagonal condition with the set $N_4(p) \cap N_4(q)$).

(c) State the unique path that remains after applying m-adjacency, and explain how this resolves the ambiguity present in 8-connectivity.

8. (Distance Metric Computation)
For two pixels $p = (0, 0)$ and $q = (3, 4)$ in a digital image:

(a) Compute the **Euclidean distance** $D_e(p, q)$.
(b) Compute the **city-block (Manhattan) distance** $D_4(p, q)$.
(c) Compute the **chessboard distance** $D_8(p, q)$.
(d) For each metric, describe the geometric shape formed by all pixels equidistant from a central pixel (i.e. the iso-distance contour), and state the practical scenario in which each metric is most appropriate.

9. (Linearity Analysis of Image Operators)
Recall that an operator H is **linear** if and only if it satisfies both additivity and homogeneity simultaneously:

$$H\big[a\, f_1(x, y) + b\, f_2(x, y)\big] = a\, H[f_1(x, y)] + b\, H[f_2(x, y)]$$

For each operator below, determine whether it is **linear or nonlinear**. For nonlinear operators, identify the specific condition (additivity or homogeneity) that fails and provide a counterexample.

(a) $H[f(x, y)] = 2f(x, y)$

(b) $H[f(x,y)] = \big[f(x,y)\big]^2$
(c) $H[\mathbf{f}] = \max\{\mathbf{f}\}$ (maximum of all pixel values in the image)
(d) $H[\mathbf{f}] = \text{median}\{\mathbf{f}\}$ (median of all pixel values in the image)
(e) $H[\mathbf{f}] = \mathbf{f} \odot \mathbf{f}$ (element-wise self-multiplication of the image)

10. (Connected Components and Region Analysis)
Consider the following 5×5 binary image, where foreground pixels (value = 1) are labelled A through E:

0	0	0	0	0
0	1_A	1_B	0	0
0	1_C	1_D	0	0
0	0	0	0	0
0	0	0	0	1_E

Using **8-connectivity** for the foreground ($V = \{1\}$):

(a) Identify all **connected components** in the image and list the pixels belonging to each.
(b) Identify the **regions** present, state whether they are adjacent or disjoint, and define the foreground and background.
(c) Determine the **boundary** of each region (pixels inside the region with at least one neighbour outside it).
(d) Determine whether any **holes** exist in this image. Justify your answer using the formal definition.